

Example function `Plate(DAC, PixReg)`

- DAC will be a 6-bit DAC value (0 to 31) that sets the value of the plating current DAC. Full scale is ~200 nA
- PixReg is the value that is written to each pixel register. To turn plating on for all channels, it can be the same value. See the table below: the pixel register is

```
+--+--+--+--+--+--+--+
|0|0|D|U|0|G|G|G|
+--+--+--+--+--+--+--+
```

Where G is the gain (doesn't matter for plating, but we've been using gain=20, or GGG=011).

To turn plating ON:

If U=1, then current will be sourced (from a compliance voltage of +1V)

If D=1, then current will be sunk (from a compliance voltage of -1V)

- Do not set both U and D to 1 at the same time

To turn plating OFF:

Set U=0 and D=0.

For example, start plating@10 nA: `Plate(2, 0x13)`

Stop plating: `Plate(2, 0x03)`

Pseudocode:

```
reg_value 0x1B000 + DAC  -- BiasPlate
reg_value 0x1B244        -- address DAC register
reg_value 0x1B308        -- write register

reg_value 0x1B000 + PixReg - PixReg
for(r=0; r<16; r++){
    for(c=0; c<31; c++){
        reg_value 0x1B100+r    -- row
        reg_value 0x1B200+c    -- col
        reg_value 0x1B308      -- write register
    }
}
```

Registers

e-chip 19 has 16 Rows x 32 Columns. When ROW = 0x00 – 0x0F and COL = 0x00 – 0x1F data are written to the individual pixel registers. Otherwise, when COL is greater than 0x3F (ROW = don't care) data are written to other registers:

Column (Binary)	Registers
0b000x xxxx	Pixel
0b001x xxxx	Reserved
0b010x xxxx	DACs
0b011x xxxx	Clock Patterns, Control Reg
0b1xxx xxxx	Reserved

DACs

DACs

T	C	D	
1	0	--	Write data
1	2	0x4n	Write "column" (address n=0..F)
1	3	0x08	Write to register

n	DAC	n	DAC
0	BiasK1	8	BiasBuf
1	BiasK2	9	Spare (to pad)
2	BiasLVM	A	BiasColDrive
3	LVDSRx	B	LVDSTx
4	BiasPlate	C	BiasLVP
5	BiasK3	D	BiasVC
6	BiasAmp	E	BiasAAF
7	BiasCMP	F	BiasOff

DACs are 5 bits.

Reset value for all DACs = 0x08

Pixel Registers

T	C	D	
1	0	--	Write data
1	1	0xr	Write row (r=0x00 .. 0x0F)
1	2	0xc	Write column (c=0x00 .. 0x1F)
1	3	0x08	Write to register

D bit	Function	RST
0	Gain	001
1	0:G=200 1:G=100 2:G=50	
2	3:G=20 4:G=10 5:G=5	
3	CLMP	1
4	Plate UP ¹	0
5	Plate DN	0
6	Reserved	0
7	Reserved	0

¹Erratum: 20200720 (UP not UPb)